Paleomagnetism of the Columbia Plateaus

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Abstract. The natural remanent magnetism (NRM) in Miocene lavas of the Columbia plateaus was surveyed in two stages corresponding to a natural subdivision of the plateaus into the Oregon plateaus and Owyhee upland in the south, and the Columbia River area in the north. In general, the lavas of the southern area are magnetically stable, and large sections often feature only a single magnetic polarity. In the northern area, sections of similar thickness are divided into several successive polarity zones and magnetically unstable lavas are common. By comparison with the inferred lava extrusion rate and the grouping of NRM directions in a well-defined transition from normal to reversed magnetic polarity across 19 lavas, stratigraphically separated groups of NRM directions in two other single sections in the southern area are interpreted to represent pauses during otherwise relatively frequent extrusions of lava. Alternatively, these observations may be explained by sudden movements of the geomagnetic field superimposed on a regular rate of lava extrusion. The NRM for the survey as a whole shows a striking relationship to a gigantic structural feature. The NRM of the Miocene lavas of the Columbia plateaus is thus consistent with previous suggestions of a tectonic origin for the broad structural pattern of the Pacific Northwest, and further implies that much of the associated rotation has taken place since the Miocene period.

INTRODUCTION

The results of two large-scale surveys of the paleomagnetic properties of the Miocene lavas of the Columbia plateaus are described. The Columbia plateaus—one of the world's largest accumulations of lava—extend into five states (insets, Figures 1 and 11). The origin of the broad structural features of this part of the United States, particularly the configuration of the Nevadan structural zone, which involves rocks of entirely pre-Tertiary age [*Gilluly*, 1963, p. 146], is a subject of controversy. The inset in Figures 1 and 11 shows the relation of the areas surveyed to the limits of the Nevadan structural zone. *Carey* [1958] believes that the shape of this structural zone may be of tectonic origin and has applied the term orocline to such features, which he has integrated into a study of global tectonic patterns. Earlier, *Becker* [1934] suggested that many of the structural features of the western United States are due to tension in the crust arising from a clockwise torque of the entire western part of the continent. *Wise* [1963] has summarized much diverse data to support the ideas of Carey and Becker, but *Wilson* [1954] and others believe that the configuration of the Nevadan structural zone is of primary origin. *Bodvarsson and Walker* [1964] have recently presented much geological and geophysical evidence for crustal spreading in Iceland, which is entirely volcanic, and have suggested that consideration be given to the possibility of similar phenomena in the other great lava accumulations of the world, such as the Columbia plateaus.

Waters [1962, p. 158] has drawn attention to the threefold division of the Tertiary volcanic province of the Pacific Northwest. Two of the provinces are separated by the east-west Ochoco–Blue Mountains axis (Figure 1, inset), which is a complex tectonic zone of faulted and folded early Tertiary, Mesozoic, and Paleozoic rocks. In tectonic, physiographic, petrological, and, as will be described, paleomagnetic properties, the Columbia River area north of the axis contrasts with the Oregon plateaus of the Basin and Range province south of the axis.

It is convenient to present the paper in three parts. The first two parts are devoted to presentation of data from the separate surveys. The data are combined for interpretation in part 3.

PART 1. THE OREGON PLATEAUS AND OWYHEE UPLAND

Sampling Details

The Oregon plateaus are part of the Basin and Range province. Ten sections (*SA* to *SJ*,

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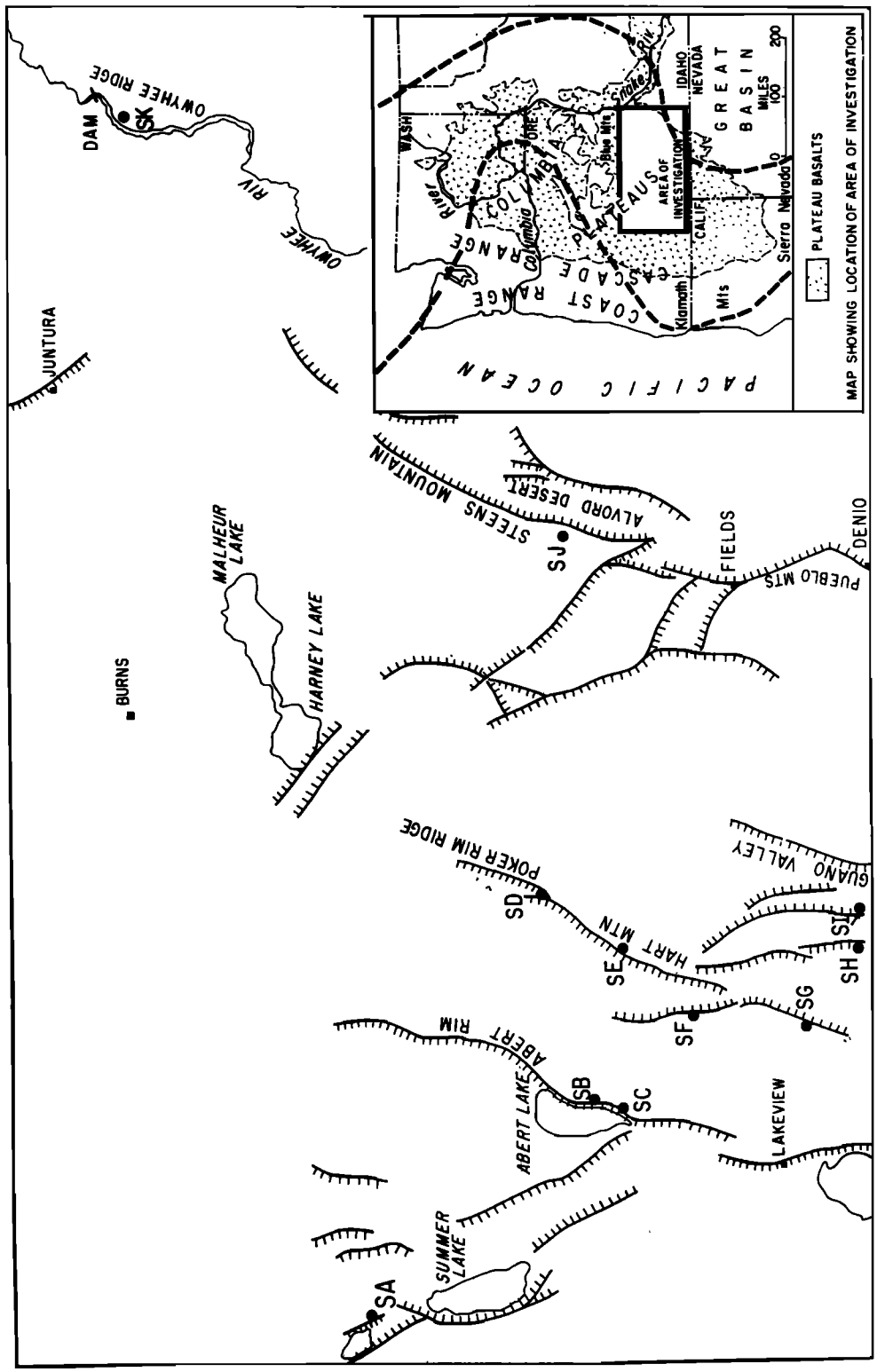


Fig. 1. Map showing locations of sections sampled and major faults in the Oregon plateaus and Owyhee upland. Inset illustrates the relation of area of investigation to the Nevadan tectonic zone.

Figure 1) have been sampled in this part of the southern Columbia plateaus, and one section (*SK*, Figure 1) was taken from the Owyhee upland, which borders the eastern side of the Basin and Range area in southern Oregon. Precise locations of each section are included in Table 1.

Baldwin [1959] summarized the geology of the area shown in Figure 1, but in doing so revealed the fact that little detailed mapping has been done. G. W. Walker (U. S. Geological Survey) has generously made available details of the central part of the area before publication.

The region consists of a vast sequence of Tertiary silicious and tuffaceous sediments containing thick sections of basalts. The area is dominated by northerly trending major faults which are still active. Sections *SG* and *SH* are petrologically the most diversified of the sections sampled, containing some thick welded and nonwelded tuffs in addition to basalts and andesites. Section *SE* consists of alternating olivine basalts and coarse pyroclastic rocks. The other sections were taken from the thick basalt sequences which contain only minor amounts of tuffaceous sediments. The most common basalt is the Steens basalt [*Fuller*, 1929], which is a porphyritic diktytaxitic basalt. Abundant Middle (?) to Late Miocene mammalian fauna correlative with the Mascall [*Downs*, 1956] and Skull Springs [*Gazin*, 1932] fauna occur locally in fine-grained tuffaceous sediments above the Steens Mountain section. Age limits of 14.5 and 21.3 m.y. for the thick Steens Mountain section were recently included in a broad radiometric dating study by *Evernden and James* [1964]. Walker (personal communication) believes that none of the lavas sampled is likely to be older than the Middle Miocene and that it is possible that the lavas in section *SF* may be younger than Upper Miocene, but no younger than Lower Pliocene.

Lavas in the area average between 3 and 5 m in thickness, compared with an average of from 15 to 25 m in the Columbia River lavas [*Waters*, 1961] north of the Ochoco–Blue Mountains axis. Thus over 150 lavas are exposed on at least two of the major scarps, which reach a maximum elevation of close to 3000 m above sea level at section *SJ*.

Field and laboratory methods. A total of 515 cores were drilled from 235 lavas in the area. In addition, 20 cores were taken from a total of 6 baked zones. The number of lavas in each section is included in Table 1. Cores, which average 12 cm in length and are 2.5 cm in diameter, were taken by means of an air-cooled

TABLE 1. Locations and Details of Sections from the Oregon Plateaus and Owyhee Upland

Section No.	Name of Section	Section Location		Approx. Vertical Extent, m	Number of Lavas Sampled
		West Longitude	North Latitude		
SA	Squaw Butte	120°48′	43°03′	150	29
SB	Abert Rim	120°10′	42°34′	240	16
SC	Abert Rim (south)	120°11′	42°32′	300	40
SD_1	Poker Jim Ridge	119°44′	42°35′	200	21
SD_2	Poker Jim Ridge	119°44′	42°35′	90	13
SE	Warner Mountain	119°45′	42°31′	120	14
SF	Plush	119°54′	42°36′	30	9
SG	Adel	119°45′	42°11′	300	25
SH_1	Greaser Canyon	119°42′	42°06′	140	12
SH_2	Greaser Canyon	119°42′	42°06′	50	4
SJ	Guano Valley	119°27′	42°06′	150	14
SJ	Steens Summit	118°33′	42°40′	150	22
SK	Owyhee Reservoir	117°17′	43°37′	300	16

gasoline-powered drill [*Doell and Cox,* 1964]. The cores were drilled at various azimuths and inclinations, one motive being to eliminate a possible effect from systematic bias during later laboratory measurements. The cores were oriented before being separated from the outcrop with a nonmagnetic orienting tool [*Doell and Cox,* 1964]. All cores were uniquely oriented in geographic coordinates to an accuracy of about 2½°.

The cores were sliced into cylinders 2.28 cm long. The NRM was measured with a spinner magnetometer [*Doell and Cox,* 1964]. The data were reduced by graphical methods.

Unstable components were examined and then removed by alternating magnetic fields, as described by *As and Zijderveld* [1958], by means of a 3-axis tumbler of the type described by *Doell and Cox* [1963]. The demagnetization experiments permitted examination of the coercivity spectrum of the NRM for representative specimens. Such spectrums are obtained by subjecting the specimens to progressively higher alternating magnetic fields and, as is becoming conventional, by calculating the fraction of the original intensity of magnetization remaining after the treatment. At frequent stages in the procedure the specimens were reoriented in the specimen holder and the experiments were repeated. The specimens were turned through 180° before reapplication of the demagnetizing field in order that an estimate could be made of the degree of response of the specimen to imperfection in the 3-axis demagnetization apparatus. If the ratio J/J_0 (where J is the intensity of magnetization after application of the given magnetic field, and J_0 is the original intensity of magnetization) differs considerably after reorientation of the specimen and reapplication of the given alternating magnetic field, an anhysteretic magnetization component [*Patton and Fitch,* 1962] may have been added to the specimen. In such cases the mean J/J_0 value and mean direction of NRM will tend to average out such spurious components added during demagnetization [*Cox and Doell,* 1963].

Results

Original magnetization. Details of the original magnetizations are not included in this paper. It is appropriate to mention, however, that the data in each section generally feature tight grouping of directions.

Unstable components. A total of 73 specimens, representing at least 4 lavas in each section, were subjected to progressive demagnetization in alternating magnetic fields ranging from 25 to 800 oe. Six representative examples of the demagnetizing process are shown in detail in Figures 2 and 3. Lack of movement of the direction of the NRM, together with the character of the J/J_0 versus H curves, is indicative of high magnetic stability.

Table 2 contains data on the J/J_0 values from about half of all experiments performed. Both values are presented when the procedure was repeated at any given magnetic field. With few exceptions, a highly stable NRM is indicated. Lavas in section *SE* showed some tendency to exhibit low coercive force properties, and in section *SI* a zone of eight successive lavas was found to have unusually low coercive force spectrums. As a result, no reliable direction of NRM could be determined from these lavas, and they are excluded from presentation here.

The J/J_0 versus H curves are typically those expected in rocks having thermal remanent magnetism (TRM), although it is not impossible that the observations are due to chemical remanent magnetism (CRM), which *Kobayashi* [1959] has shown to be indistinguishable from TRM in lavas on the basis of such curves. *Watkins and Wilson* [1964] have shown that exsolved titanium-rich lamellas in titanomagnetite exist in many of the lavas; therefore a CRM of the type mentioned by *Stacey* [1963, p. 103] may be present.

The magnitude of the alternating magnetic field required to remove unstable components was determined from the results of the progressive demagnetizing experiments. All specimens were then subjected to treatment in an appropriate alternating magnetic field as given in Table 2.

Natural remanent magnetism. With the exception of the previously mentioned highly unstable zone in section *SI*, Figures 4 to 9 illustrate the directions of NRM in all sections after treatment in the appropriate alternating magnetic fields. The demagnetization process has resulted in some decrease in scatter of NRM directions in each section. Sections *SD* and *SH* exhibit the division of NRM into two well-de-

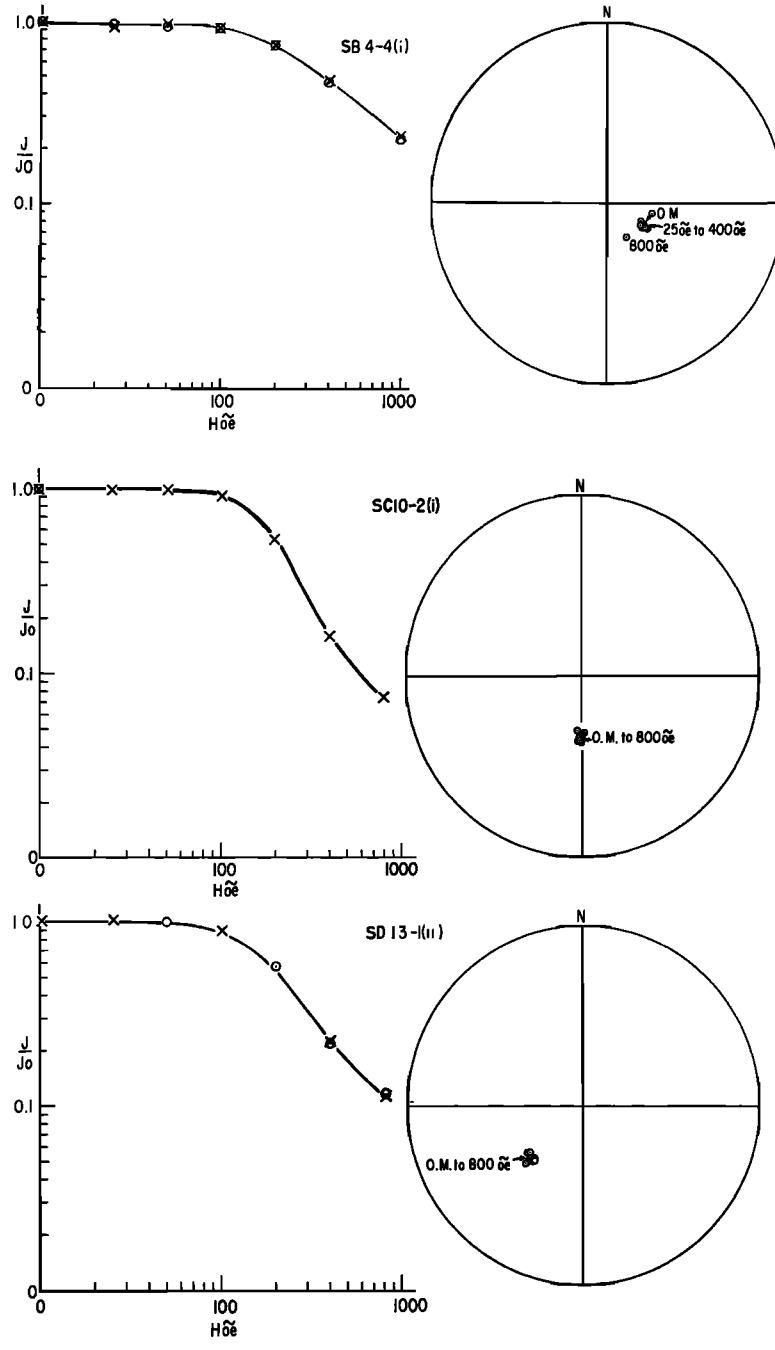


Fig. 2. Response of samples from sections *SB, SC,* and *SD* to alternating magnetic fields. Convention on graphs: J_o is the original intensity of magnetism, J is the intensity of magnetization remaining after application of an ac field of amplitude H oe, $\times$ represents the initial orientation of long axis of specimen into the demagnetizing apparatus, and $\odot$ represents the initial orientation of long axis of specimen out of the demagnetizing apparatus. Convention on the equal area projection: N is geographic north; $\odot$ represents north-seeking polarity upward; $\oplus$ represents north-seeking polarity downwards. O.M. is the original magnetization. Numbers refer to magnitude of H used. $+$ and $-$ correspond to $\times$ and $\odot$ on graph. The sample number convention is fourfold, being in the sequence: section, lava, core, specimen.

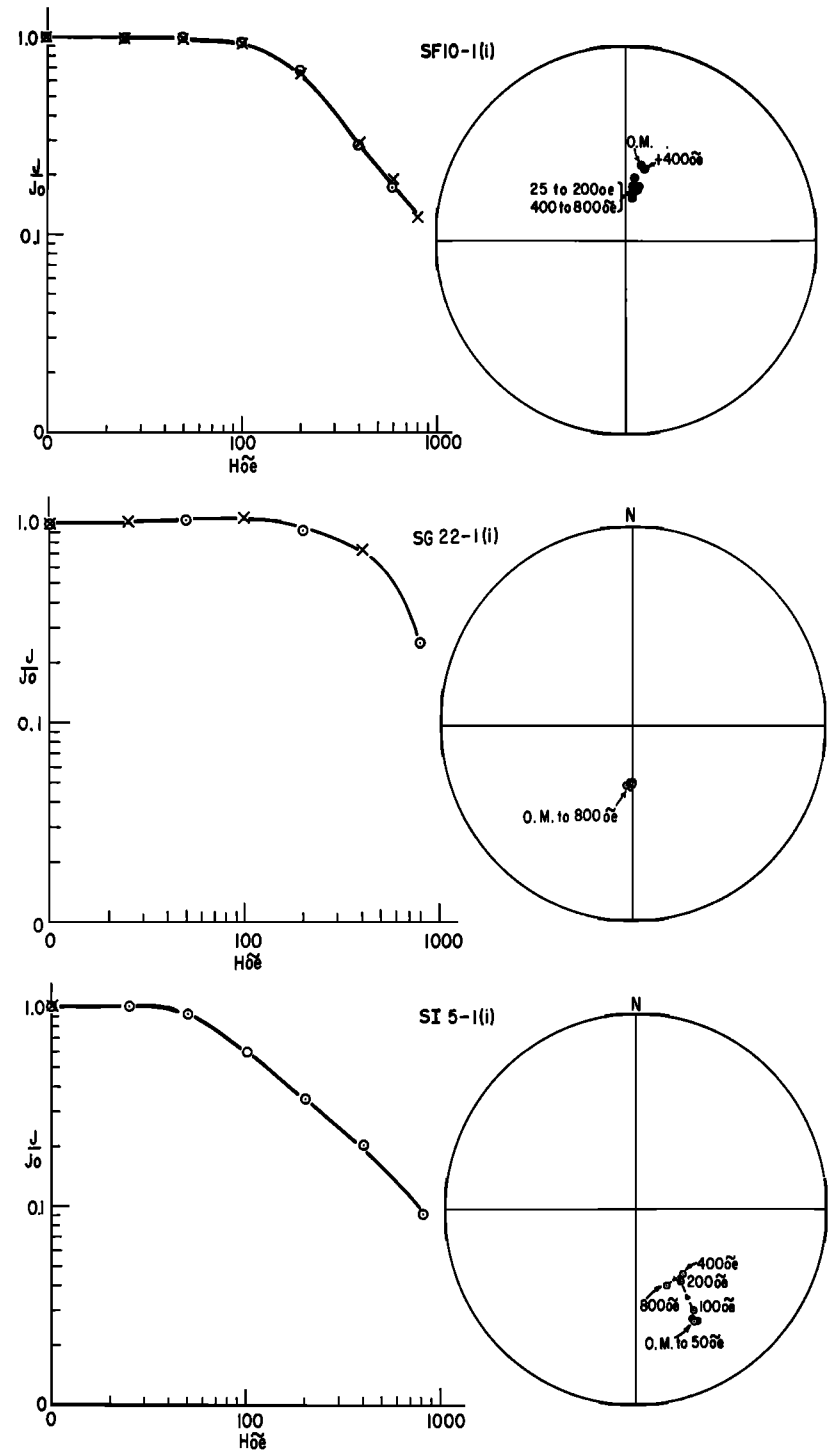


Fig. 3. Response of samples from sections *SF*, *SG*, and *SI* to alternating magnetic fields. For explanation of symbols see Figure 2.

TABLE 2. The Oregon Plateaus and Owyhee Upland Demagnetization Data

Section No.	Lava No.	Core No.	Specimen No.	J/J_0 at Following Demagnetizing Fields*						Demag. Field Chosen for Section
				25 oe	50 oe	100 oe	200 oe	400 oe	800 oe	
SA	11	2	(i)	0.999	0.990	0.873	0.600	0.198	0.088	
SA	28	2	(i)	1.020	0.995	1.010	0.722	0.294	0.201	50 oe
				0.958	0.995	0.929	0.715	0.306	0.226	
SB	6	6	(i)	0.978	0.955	0.910	0.695	0.180	0.067	
				0.970	0.970	0.910	0.685	0.181	0.069	
SB	10	2	(i)	1.010	1.010	0.935	0.815	0.433	0.191	
				1.010	0.975	0.990	0.800	0.436	0.188	
SB	16	5	(i)	1.030	1.030	1.010	0.870	0.479	0.216	50 oe
				1.040	1.030	1.010	0.860	0.510	0.222	
SB	18	3	(i)	1.020	1.000	0.975	0.790	0.419	0.234	
				0.990	1.020	0.975	0.784	0.409	0.202	
SC	10	2	(i)	0.999	0.980	0.532	0.532	0.160	0.074	
SC	37	1	(i)	0.986	1.030	1.025	0.934	0.480	0.187	50 oe
				1.030	1.030	1.025	0.930	0.492	0.180	
SD_1	8	1	(ii)	1.01	1.005	0.998	0.850	0.494	0.154	
								0.488	0.285	
SD_1	13	1	(ii)	1.01	0.994	0.888	0.573	0.226	0.113	50 oe
								0.221	0.116	
SD_2	29	1	(i)		1.240	1.230	0.935	0.460	0.183	
SD_2	36	1	(i)		1.020	0.930	0.661	0.414	0.286	
								0.410		
SD_2	38	1	(i)		1.140	1.200	1.030	0.530	0.163	
									0.152	
SD_2	40	1	(i)	1.200	1.210	1.140	0.785	0.205	0.136	
				1.200	1.210	1.120	0.775	0.305	0.142	
SE	1	2	(i)	0.974	0.894	0.693	0.457	0.199	0.117	
SE	9	1	(i)	0.771	0.603	0.588	0.338	0.231	0.115	
				0.795	0.603	0.451	0.425	0.224	0.096	
SE	10	1	(i)		0.810		0.381	0.393	0.152	50 oe
					0.810		0.490	0.267	0.269	
SE	14	2	(i)		0.970		0.702	0.444	0.277	
					0.945		0.726	0.487	0.327	
SF	9	2	(i)	0.985	1.020	0.990	0.642	0.353	0.208	
SF	10	1	(i)	0.980	0.972	0.912	0.668	0.290	0.119	200 oe
				0.985	0.978	0.918	0.661	0.282	0.056	
SG	5	1	(i)	0.977	0.937	0.960	0.520	0.304	0.092	
SG	12	2	(i)	1.010	0.990	0.967	0.622	0.376	0.184	
				1.050	0.070	0.958	0.620	0.418	0.154	200 oe
SG	16	1	(i)	1.030	1.060	1.035	0.915	0.735	0.251	
SH_1	2	1	(i)	1.000	1.040	0.970	0.658	0.350	0.164	
				1.030	1.030	0.955	0.710	0.375	0.178	
SH_1	8	2	(i)		0.990		0.900	0.600	0.275	
					1.010		0.910	0.573	0.161	50 oe
SH_1	20	1	(i)		0.967		0.558	0.281	0.182	
							0.558	0.225	0.130	

TABLE 2. Continued

Section No.	Lava No.	Core No.	Specimen No.	J/J_0 at Following Demagnetizing Fields*						Demag. Field Chosen for Section
				25 oe	50 oe	100 oe	200 oe	400 oe	800 oe	
SH_2	14	1	(i)	1.040	1.020	0.848	0.478	0.146	0.107	
SH_2	15	1	(i)		0.980		0.629	0.294	0.146	
							0.710	0.303	0.126	
SH_2	19	2	(i)		0.883		0.568	0.292	0.184	50 oe
							0.548	0.289	0.184	
SI	2	2	(i)	1.015	0.880	0.540	0.383	0.210	0.087	
SI	5	1	(i)	1.000	0.925	0.595	0.347	0.202	0.089	400 oe
SI	17	2	(i)	1.045	1.060	0.975	0.767	0.390	0.194	and
SI	22	2	(i)	0.970	0.990	0.940	0.660	0.198	0.068	50 oe
SJ	13	2	(i)	0.835	0.770	0.637	0.430	0.281	0.161	
				0.835	0.756	0.635	0.421	0.258	0.123	50 oe
SJ	19	1	(i)	0.830	0.730	0.600	0.465	0.244	0.820	
SK	1	1	(i)	1.010	1.160	0.713	0.144	0.083	0.043	
								0.083		
SK	2	2	(i)	1.000	0.642	0.113	0.019	0.004	0.004	
								0.006	0.007	200 oe
SK	10	1	(i)	1.010	0.963	0.880	0.552	0.183	0.108	
SK	16	2	(i)	1.010	0.970	0.850	0.665	0.455	0.196	

* J/J_0 is the ratio of intensity of magnetization after and before treatment in alternating field.

fined groups. Moreover, this dual distribution of NRM is stratigraphically separated within the two sections.

Section *SH* features an apparent oscillation of NRM in successive lavas between the two well-defined groups [*Watkins*, 1963*a*], but the study of petrological details in the lavas, when integrated with the paleomagnetic data, confirms suspected minor faulting in the section [*Watkins and Larson*, 1965]. This minor faulting causes repeated sampling of the same lavas, which accounts for the spurious oscillation of NRM directions. When proper account is taken of the faulting, there are only two groups of NRM directions successive in time and stratigraphically distinct. All other sections contain successive lavas with no repetition of lavas within a single section, although it is not impossible that the same lava may occur in more than one section. Directions of NRM in baked zones associated with the sampled lavas in sections *SB* and *SH* are included in Figures 4 and 7.

Statistical analysis. The data shown in Figures 4 to 9 have been analyzed according to the methods of *Fisher* [1953] and *Watson and Irving* [1957], although not all data conform exactly to the necessary density distribution [*Watson*, 1956]. Since the well-defined dual direction of NRM in sections *SD* and *SH* are stratigraphically separated, they have been subjected to separate statistical analysis, and thus analyses of 13 separate sections are presented in Table 3. The virtual geomagnetic poles are presented in Figure 10. The ovals for all but the mean virtual geomagnetic pole[2] for the entire survey are omitted for clarity, and a mean virtual geomagnetic pole, determined by *Doell and Cox* [1961*a*, p. 294] from several reliable paleomagnetic surveys of Miocene rocks elsewhere in the world, has been added to Figure 10.

[2] *Irving* [1964] prefers the term paleomagnetic pole for the surface expression of the geocentric dipole calculated from paleomagnetic data, since he feels that the calculated pole is an estimate of the ancient geographic pole. The term virtual geomagnetic pole [*Cox and Doell*, 1960] is preferred here since no reliable estimate of the position of the spin axis is implied.

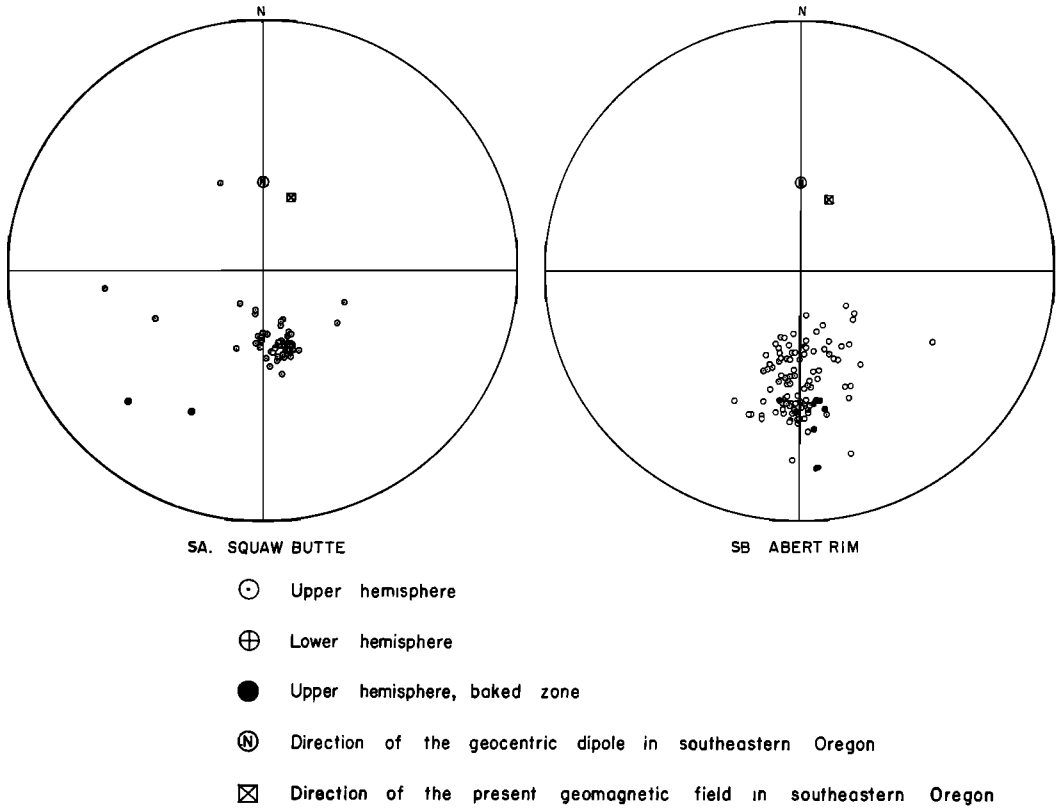


Fig. 4. Directions of NRM in all cores from sections *SA* and *SB* after treatment in an alternating magnetic field of 50 oe. Projection is equal area.

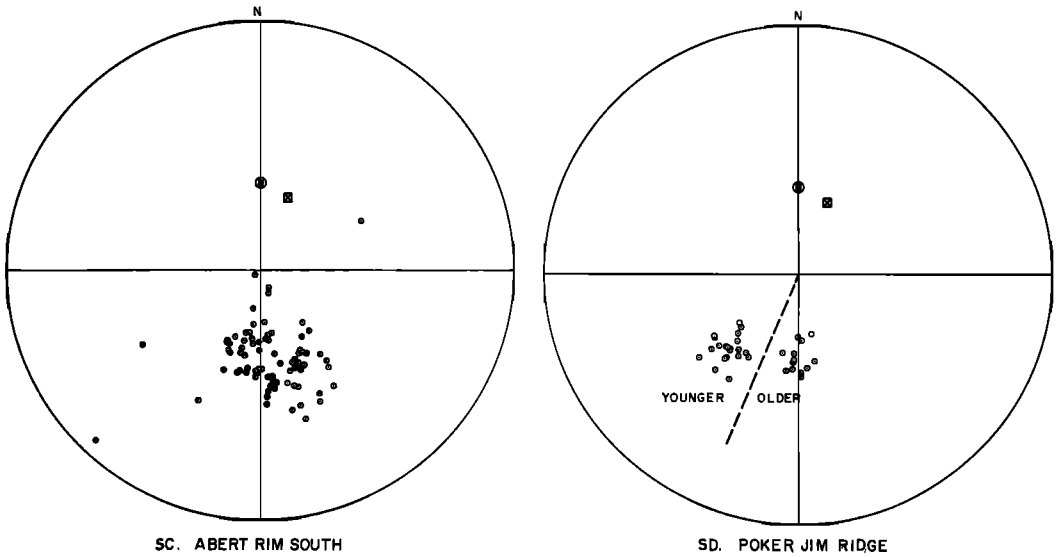


Fig. 5. Directions of NRM in all cores from sections *SC* and *SD* after treatment in an alternating magnetic field of 50 oe. Symbols as for Figure 4. Younger NRM group in section *SD* is directly above older group.

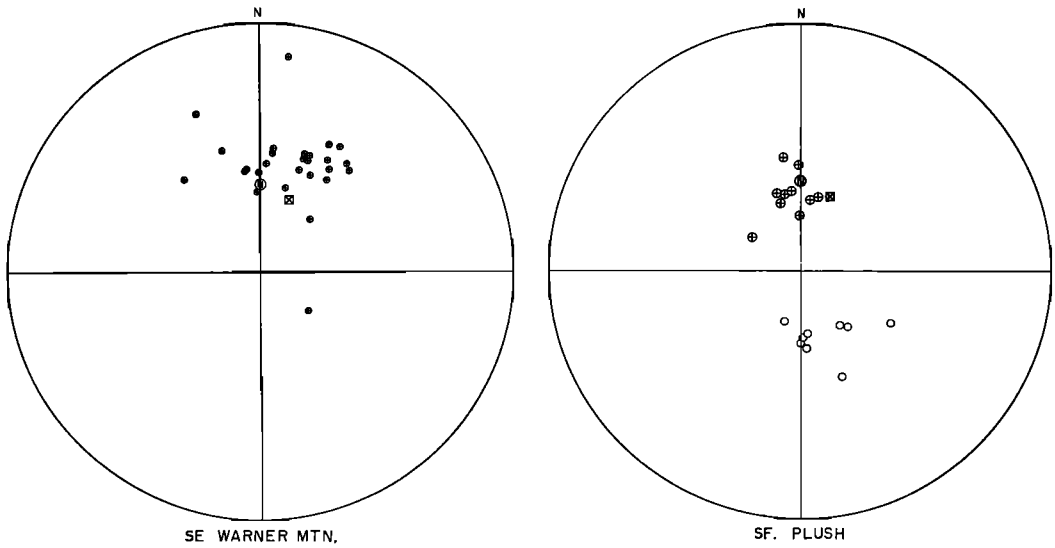


Fig. 6. Directions of NRM in all cores from sections *SE* and *SF* after treatment in alternating magnetic fields of 50 and 200 oe, respectively. Symbols as for Figure 4.

Part 2. The Columbia River Basalts

Sampling Details

The plateaus underlain by the Columbia River basalts are predominantly that part of the Columbia plateaus north of the Ochoco–Blue Mountains axis and east of the Cascade Mountains (Figure 11, inset). Ten sections (*NA* to *NJ*, Figure 11) have been sampled. The precise location of each location is included in Table 4. The area investigated is situated around the axis of the gigantic structural feature or orocline [*Carey*, 1958] illustrated in the inset of Figure 11.

A total of 198 lavas have been sampled. This number no doubt includes some repeated sam-

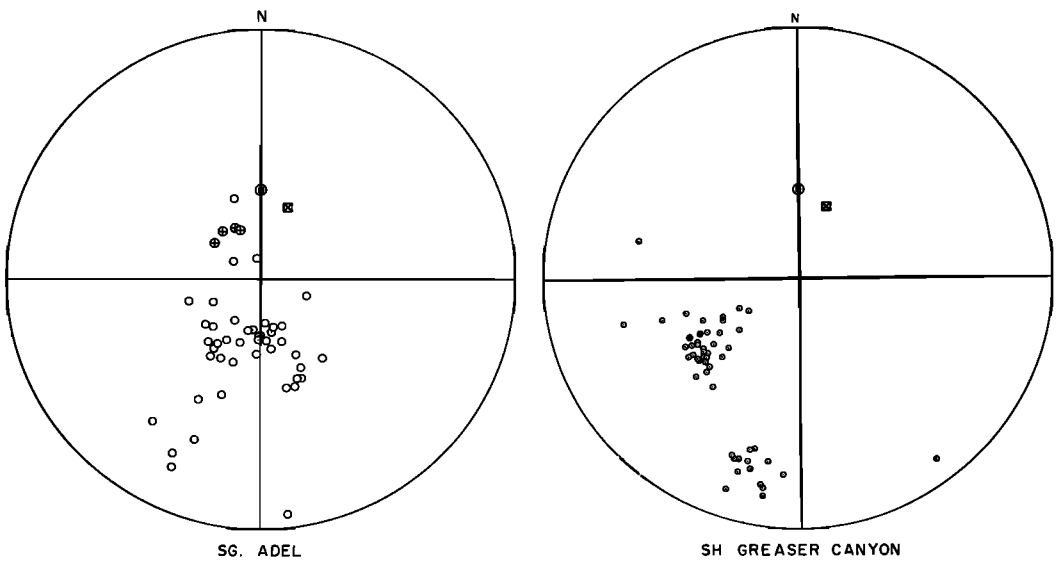


Fig. 7. Directions of NRM in all cores from sections *SG* and *SH* after treatment in alternating magnetic fields of 200 and 50 oe, respectively. Symbols as for Figure 4.

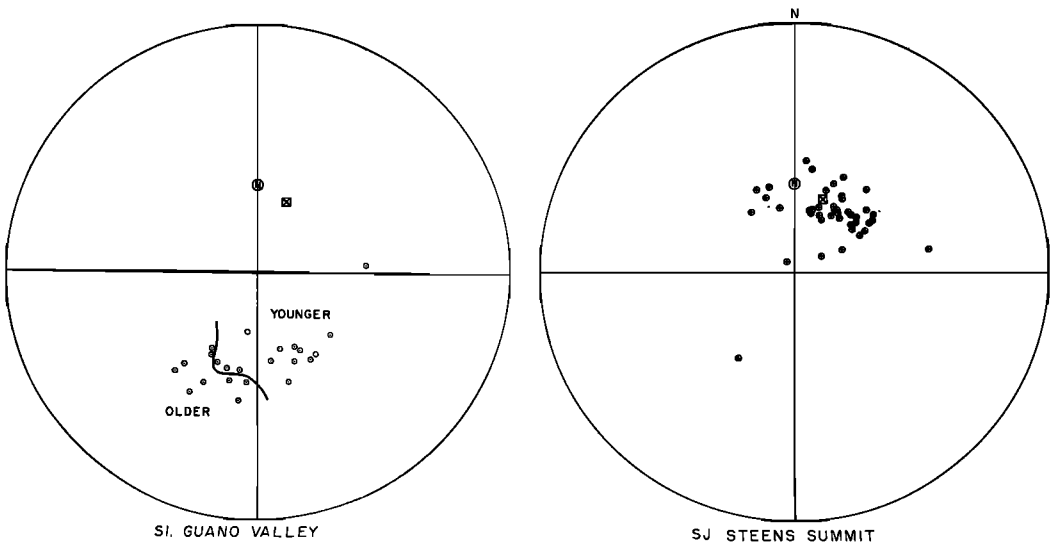


Fig. 8. Directions of NRM in all cores from section *SI*, excluding cores from a central zone of magnetically highly unstable lavas. The diagram also includes NRM from all cores in section *SJ* after treatment in alternating magnetic fields. 400-oe and 200-oe magnetic fields were used on lavas respectively above and below the unstable zone in section *SI*. Lavas younger and older than the unstable zone in section *SI* are indicated. A 50-oe magnetic field was used on the cores from section *SJ*. Symbols as for Figure 4.

pling of the same lava in different sections. The number of lavas sampled in each section is included in Table 4. In rare cases, where outcrop was limited, it is possible that the same lava was sampled more than once in the same section, and in section *ND* some local faulting may have accomplished the same condition. The nature of the outcrops, which are mainly along the steep canyons of the Snake River and its tributaries, has made possible a sampling of sections of continuous successive lavas.

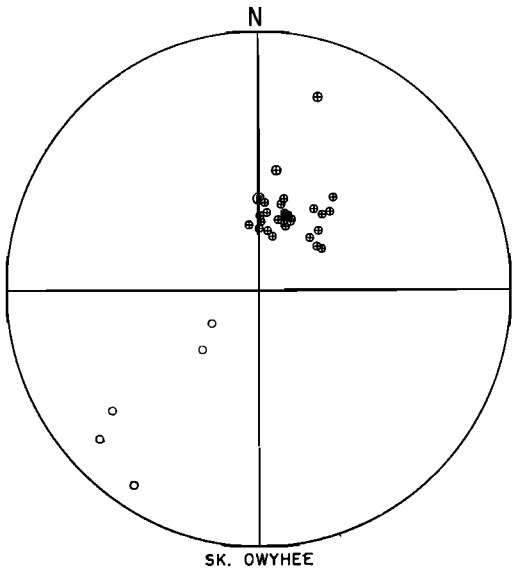


Fig. 9. Direction of NRM in all cores from section *SK* after treatment in an alternating magnetic field of 50 oe.

The sections are all in the Columbia River basalt, which *Waters* [1961] has divided into the Picture Gorge basalt and the younger Yakima basalt. The division is based on stratigraphic and petrological grounds. As part of that study, Waters examined details of the petrology in 7 of the 10 sections sampled during the paleomagnetic survey. The majority of the lavas sampled are in the Yakima basalt, which, despite its enormous volume, is superficially uniform, occurring as nonporphyritic lavas, with an average thickness of about 20 m. In one section (*NG*, Imnaha) Waters recognized both Picture Gorge and Yakima basalts, separated by a slight unconformity. A few lavas at the top of at least two sections (*NE*, Pendleton; and *NF*, Minam), resulted from later events which Waters refers to as the Lake Yakima–Ellensburg phase of extrusion. The only appreciable structure in the sections surveyed occurs

TABLE 3. The Oregon Plateaus and Owyhee Uplands Statistical Analysis of Results for Sections *SA* to *SK*

Section No.	D	I	R	N	B	α_{95}, deg (1 level)	w	β	α_{95}, deg (2 level)	δ_p, deg (1 level)	δ_m, deg (1 level)	k	ϕ'	θ'
SA	357°45′	+71°03′	27.004	54	29	7.4	43.10	18.73	6.7	11.8	12.9	14.0	233°15′	75°03′
SB	358°03′	+51°45′	15.537	98	16	6.6	43.48	60.57	5.0	6.0	9.0	32.4	68°30′	79°45′
SC	0°10′	+61°00′	38.693	79	40	4.2	19.28	122.69	4.1	4.9	6.5	30.0		90°00′
SD_1	41°54′	+56°39′	20.869	21	21	2.6				2.7	3.8	152.0	325°30′	57°45′
SD_2	0°48′	+63°07′	12.946	13	13	2.8				3.5	4.4	222.0	256°40′	87°59′
SE	17°59′	+52°58′	13.535	28	14	7.6	29.43	62.15	6.8	7.3	10.6	28.0	335°47′	73°17′
SF	346°51′	+66°08′	8.911	18	9	5.4	45.59	589.97	5.1	7.3	8.9	90.0	187°01′	78°59′
SG	9°43′	+65°57′	23.533	50	25	7.4	50.87	19.16	7.0	9.8	12.1	16.4	285°20′	80°52′
SH_1	56°25′	+51°11′	11.849	36	12	5.1	37.9	209.60	4.7	4.7	7.4	73.0	323°54′	44°42′
SH_2	13°44′	+21°44′	3.996	14	4	3.2	1059.3	1458.6	2.2	1.7	3.2	836.0	36°06′	52°54′
SI	5°48′	+59°35′	13.423	28	14	8.5	84.94	29.47	7.5	9.7	12.8	23.4	348°32′	85°20′
SJ	20°00′	+66°00′	21.599	44	22	4.3	26.51	826.5	4.2	5.8	7.1	52.5	301°23′	74°45′
SK	28°08′	+61°03′	15.173	32	16	8.9	144.1	24.34	8.6	10.5	13.6	18.2	332°43′	65°59′
All sections	13°20′	60°45′	219.14	515	235	2.3				2.9	3.8	12.6	327°47′	80°05′

All data from magnetically reversed lavas are transposed to northern hemisphere.

D, mean declination angle, east of north, assigning unit vector to each lava.
I, mean inclination angle below horizontal, assigning unit vector to each lava.
R, resultant vector, assigning unit vector to each lava.
N, number of cores.
B, total number of lavas.
α_{95}, semi-vertical angle of 95% confidence cone.
w, within-lava confidence parameter.
β, between-lava confidence parameter.
δ_p, semi axis of 95% oval of confidence along great circle between section site and virtual geomagnetic pole.
δ_m, semi axis of 95% oval of confidence, normal to δp.
k, $B - 1/B - R$ (precision parameter).
ϕ', east longitude of virtual geomagnetic pole.
θ', latitude of virtual geomagnetic pole.

in section *NJ*, which is the type section of the Picture Gorge basalt. Here thick lavas dip up to 20°.

On paleontological evidence, the age of the sections varies from Middle Miocene to Early Pliocene. At the type section, the Picture Gorge basalt is overlain by the Middle or Late Miocene Mascall formation [*Merriam*, 1910; *Chaney and Axelrod*, 1959]. The Yakima basalt overlies or is intercalated with Middle or Late Miocene flora [*Chaney and Axelrod*, 1959], which is regarded as older than the flora of the Mascall formation. The late Yakima–Ellensburg outpourings are assigned to the Early Pliocene on the basis of evidence from both vertebrata and flora [*Axelrod*, 1950]. *Evernden and James* [1964], however, have radiometrically dated rocks which include material associated with the above paleontological studies and have concluded that, in the absence of other stratigraphic criteria, ages determined from floral evidence may be subject to large error. Included in their results are dates which place approximate limits of 15.4 and 24.9 m.y. on the age of the Columbia River basalt at Picture Gorge, thus suggesting that much of the Picture Gorge basalt is in the Lower Miocene. The remainder of these recent data does not imply that the sections sampled are other than Miocene in age, and, as stated earlier, most of the lavas sampled are Yakima basalt, which is younger than the Picture Gorge basalt.

Field and Laboratory Methods

Two cores were drilled in each lava to facilitate the statistical requirements for determining a mean direction of NRM for each section [*Watson and Irving*, 1957]. A few baked interbasaltic sediments were sampled.

The collection and orientation procedure and the laboratory methods are essentially the same as described in part 1. Cores were oriented to an accuracy of 3° with respect to geographical coordinates.

Results

Original magnetization. The details of the original magnetization of all specimens are not included in this paper. Considerable scatter in NRM direction was observed. This is similar to

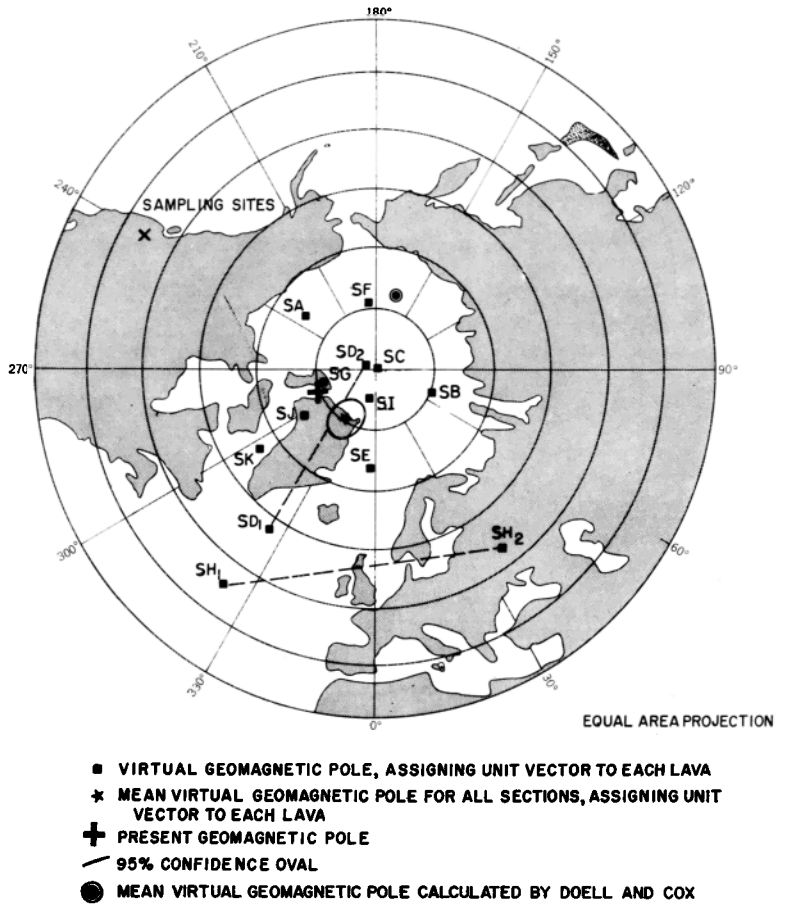


Fig. 10. Virtual geomagnetic poles for sections *SA* to *SK*. (For Doell and Cox calculated mean, see *Doell and Cox* [1961*a*, p. 294].

some of *Campbell and Runcorn's* [1956] results from a study of the NRM of a suite of lavas from the Columbia River basalts.

Both reversed and normally magnetized specimens were obtained from single lavas in the survey. This property would be particularly important in the event that local stratigraphic studies include field measurements of NRM [*Doell and Cox,* 1962] similar to those of *Einarrson* [1958] in Iceland, since a limited number of field measurements of the polarity of a given lava might obviously yield misleading data.

Unstable components. Specimens from at least four lavas in each section were subjected to progressive demagnetization in alternating magnetic fields ranging from 12.5 to 800 oe. The details of the behavior of 9 specimens in response to progressive demagnetization are shown in Figures 12 and 13. These are typical of many of the 47 specimens so examined. The data defining the J/J_0 versus H curves for most of the specimens examined are given in Table 5. These data, together with those in Figures 12 and 13, illustrate the great variation in the coercivity spectrums for the NRM of the specimens. The J/J_0 versus H curves are not typically characteristic of TRM in lavas, but TRM would be expected to be the dominant type of remanent magnetism in the lavas, after demagnetization.

Examples of pseudo-normally magnetized specimens are included in Figures 12 and 13. Before removal of the unstable components these samples have a normal polarity, often in nearly the same direction as the present geomagnetic field. Similar low-coercivity basalts in Australia

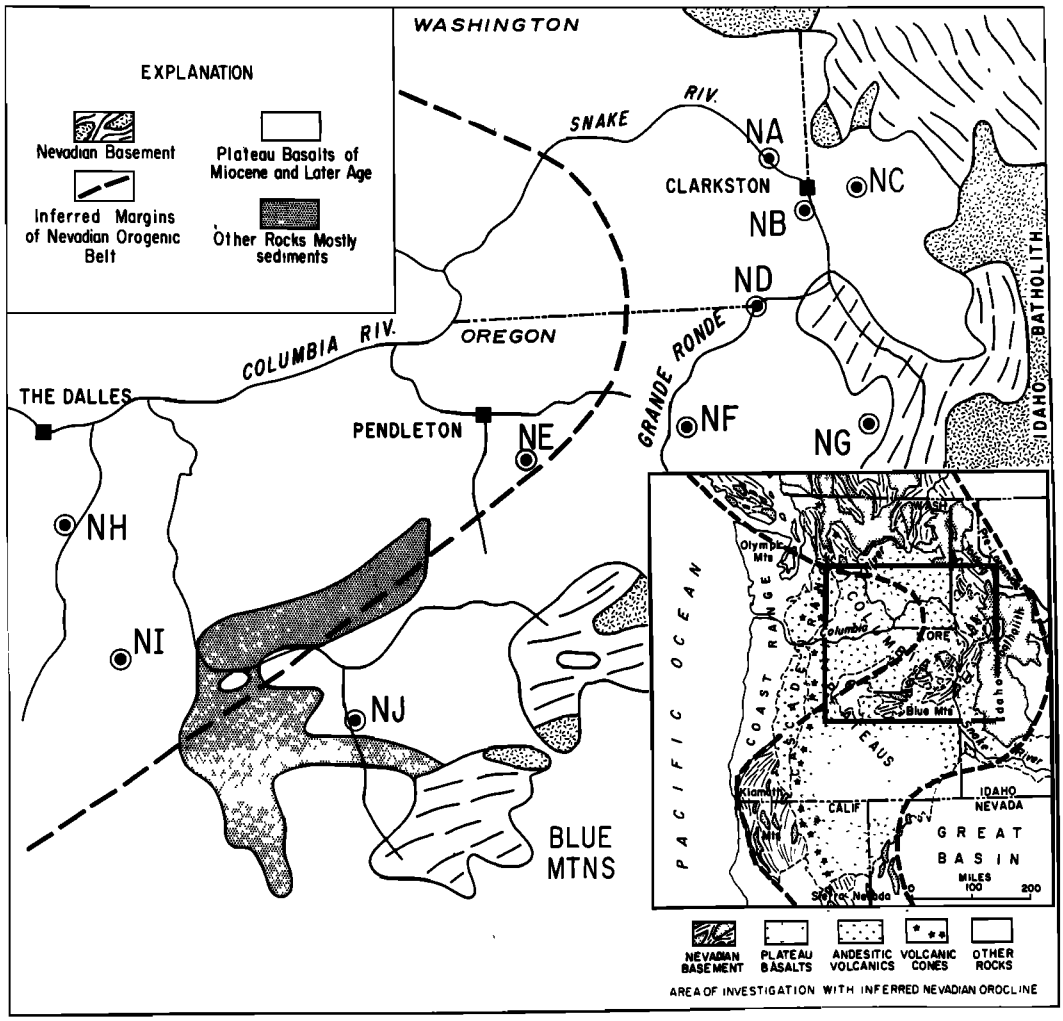


Fig. 11. Map showing locations of sections sampled in the Columbia River basalt. Inset illustrates relation of area of investigation to the Nevadan tectonic zone.

have been described by *Mumme* [1962]. These very large viscous components are typical of many of the lavas encountered and are among the properties which contrast with properties of the lavas of the area south of the Ochoco–Blue Mountains axis, described in part 1, where virtually all the lavas sampled possess very small unstable components.

An alternating magnetic field of 200 oe, which was considered adequate to remove viscous components in the lavas, was applied to specimens from all cores. This testament showed that the mixed polarities of the original magnetization in some lavas was owing to stability differences.

Natural remanent magnetism. Figures 14 to 18 show the NRM from all cores, following the application of a 200-oe alternating magnetic field. A considerable decrease in scatter of NRM directions has resulted, although sections *ND* and *NI* by no means have well-defined groups of NRM. Removal of viscous components from the lavas has resulted in the polarity of the few baked interbasaltic specimens being the same as the associated overlying lavas.

Statistical analysis. The statistical methods of *Fisher* [1953] and *Watson and Irving* [1957] have been applied to all the data in Figures 14 to 18, and the results are presented in Table 6.

TABLE 4. Section Locations and Details

Section No.	Name of Section	Location of Sections		Approx. Vertical Extent, m	No. of Lavas Sampled
		West Longitude	North Latitude		
NA	Snake River, Whitman County	117°15′	46°31′	530	22
NB	Asotin	117°02′	46°20′	370	10
NC	Lapwai Canyon	116°36′	46°14′	340	16
ND	Grande Ronde, Asotin County	117°12′	46°05′	490	30
NE	Pendleton	118°34′	45°38′	240	18
NF	Minam, Union County	117°45′	45°35′	340	22
NG	Imnaha, Wallowa County	116°47′	45°32′	850	36
NH	Tygh Valley, Wasco County	121°10′	45°17′	270	19
NI	Cow Canyon	120°54′	44°51′	200	10
NJ	Picture Gorge	119°37′	44°31′	200	15

As in part 1, the necessary density distribution of the data is not ideally approximated in all sections. The virtual geomagnetic pole for each section is shown in Figure 19. The mean virtual geomagnetic pole derived from other surveys is included. The 95% confidence oval for the mean virtual geomagnetic pole for the entire survey of the Columbia River basalts is included. Other 95% confidence ovals are omitted for clarity.

Part 3. Interpretation of the Data

Polarities. Eighty per cent of the lavas examined in the southern area are reversely magnetized. Only 2 of the 13 sections in the south contain both normally magnetized (N) and reversely magnetized (R) lavas, and up to 40 successive lavas have been observed to have a single polarity.

In the northern area, however, all but one of the sections have both N and R lavas. The thickest section (section *G*, Imnaha) has five successive pairs of N and R zones. N and R lavas occur in approximately equal numbers in the northern area.

A high between-lava confidence parameter (Tables 3 and 6) for a section indicates a small variation of mean NRM direction between lavas. This can be caused either by a rapid extrusion of lavas during normal variation of the Miocene geomagnetic field or by very slow variation of the Miocene geomagnetic field during a period of less rapid extrusion of the lavas. The value of the between-lava confidence parameter (β) in the south varies from 18.3 to 1059.3, with a median value of 43.48. In the north the between-lava confidence parameter varies from 6.29 to 38.64, with a median value of about 22.5. This large difference in the relative values of the between-lava confidence parameter suggests that the lava extrusion rate in the southern area was more rapid than in the northern area, assuming that the respective ambient Miocene magnetic fields had similar secular variation rates in both areas. A more rapid extrusion rate in the south is also consistent with the difference in polarity variation between the two areas, although it is not certain that the polarity epochs [*Cox et al.*, 1963] encountered in the north are the same duration as those represented in the south. The absolute dating which has been done [*Everden and James*, 1964], however, suggests that the lavas in the southern area are of a similar age to at least some lavas in the northern area, and no obvious hiatuses which might exclude a dif-

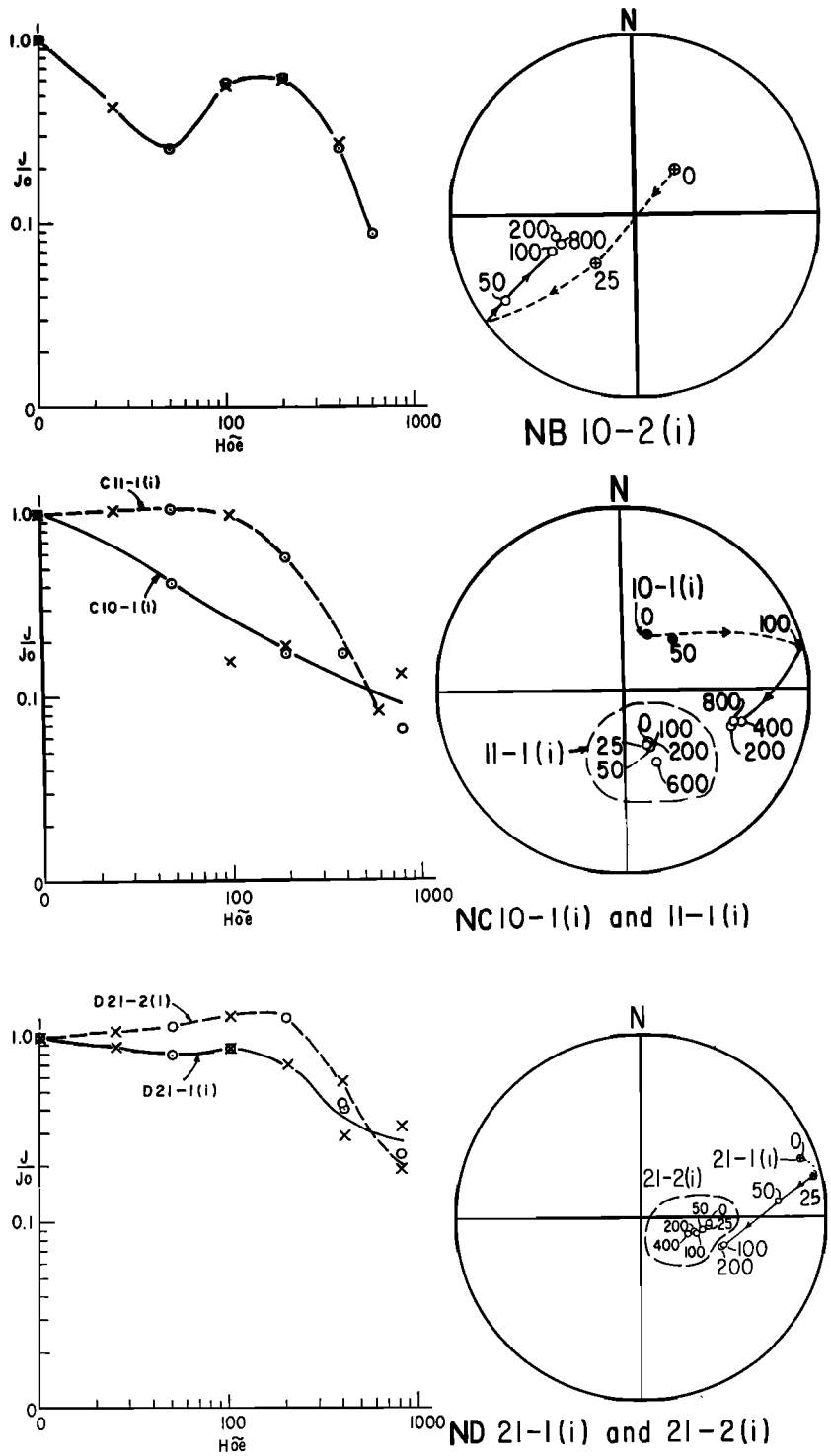


Fig. 12. Response of samples from sections *NB, NC,* and *ND* to alternating magnetic fields. Symbols as for Figure 4.

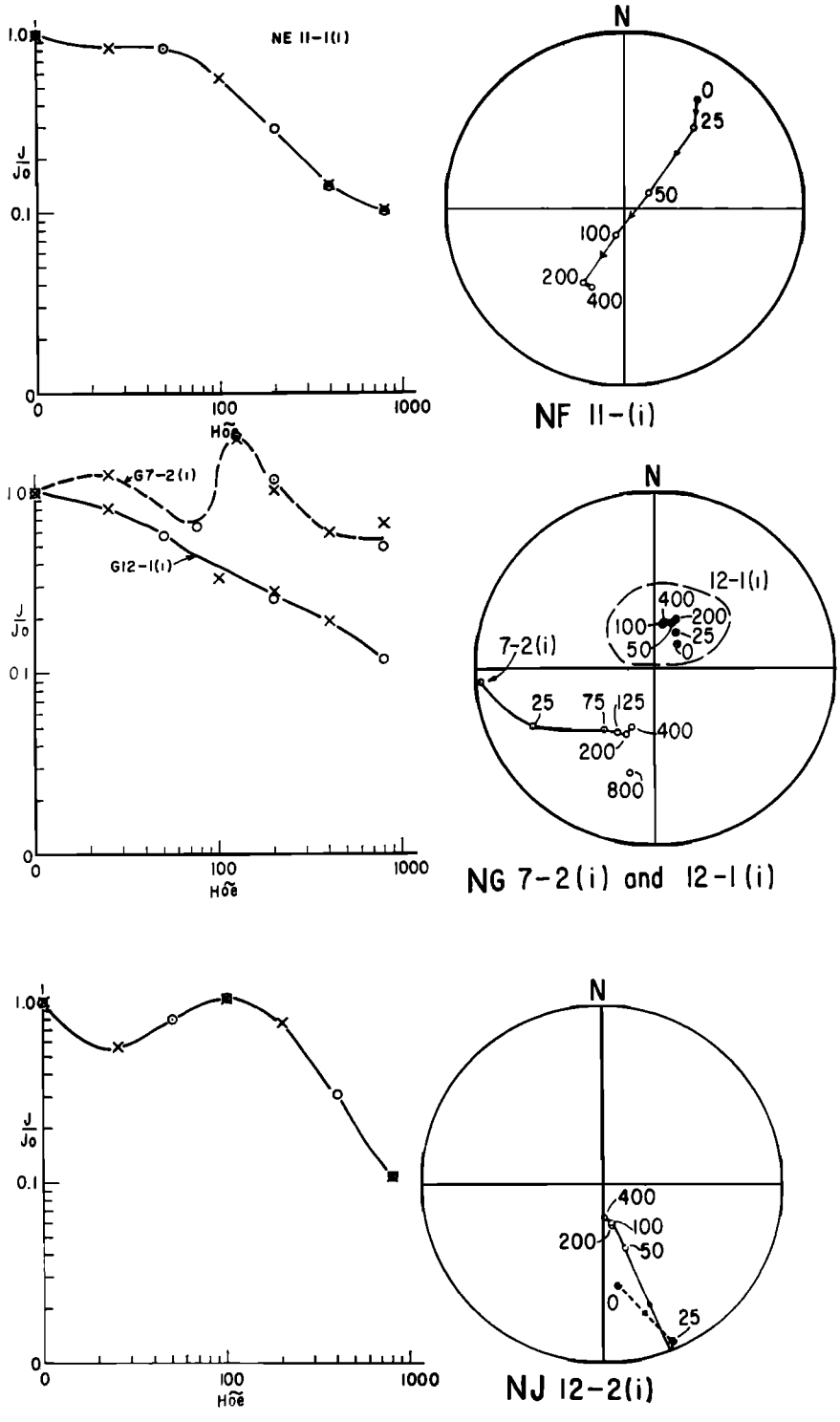


Fig. 13. Response of samples from sections *NF*, *NG*, and *NJ* to alternating magnetic fields. Symbols as for Figure 4.

TABLE 5. Columbia River Basalts Demagnetization Data

Section No.	Lava No.	Core No.	Specimen No.	J/J_0 at Following Demagnetizing Fields						
				12.5 oe	25 oe	50 oe	100 oe	200 oe	400 oe	800 oe
NA	2	1	(i)	0.955	1.320	2 630	2.150	1.530	0.705	0.572
								1.500		
NA	7	2	(i)	0.872	0.964	0.670	0.189	0.093	0.059	0.154
										0.023
NA	11	2	(i)	1.020	1.000	0.990	0.960	0.838	0.616	0.412
NA	22	1	(i)	0.915	0.588	0.258	0.147	0.061	0.017	0.008
									0.016	0.009
NB	5	2	(i)			0.868	0.740	0.364	0.190	0.146
NB	6	2	(i)		0.960	1.080	1.180	0.778	0.267	0.150
NB	9	2	(i)		1.230	1.340	2.220	1.530	0.435	0.087
							2.320	1.560		
NB	10	2	(i)		0.425	0.256	0.556	0.610	0.255	0.086
							0.563	0.610	0.272	
NC	7	2	(i)	0.994	0 970	0.935	0.710	0.251	0.072	0.017
NC	10	1	(i)			0.415	0.154	0.170	0.165	0.128
								0.180		0.063
NC	11	1	(i)		1.030	1.040	0.945	0.560	0.180	
NC	12	1	(i)	1.010	0.980	0.828	0.630	0.540	0.395	0.161
ND	1	1	(i)		0.962	0.942	0.895	0.672	0.388	0.296
ND	16	1	(i)		1.010	1.070	1.080	1.050	0.795	0.413
ND	21	2	(i)		1.075	1.160	1.195	1.190	0.594	0.196
									0.434	
ND	32	1	(i)		0.760	0.538	0.363	0.333	0.300	0.260
NE	6	2	(i)		0.945	0.890	0.763	0.560	0.317	0.190
NE	16	2	(i)		1.000	1.000	0.980	0.925	0.552	0.176
NF	4	2	(i)		1.020	0.723	0.493	0.290	0.170	0.096
NF	11	1	(i)		0.825	0.824	0.560	0.293	0.140	0.101
									0.142	0.104
NF	15	2	(i)		1.330	1.340	0.845	0.442	0.265	0.182
										0.164
NF	22	1	(i)			0.575	0.304	0.093	0.050	0.049
								0.108	0.064	0.036
NG	7	2	(i)		1.250	0.780	1.450	1.170	0.585	0.496
							1.460	1.00		0.665
NG	12	1	(i)		0.812	0.585	0.348	0.264	0.197	0.120
								0.258		
NG	21	1	(i)		1.100	1.240	0.930	0.885	0.323	0.070
NG	33	1	(i)		1.040	1.110	1.110	0.830	0.390	0.334
								0.830		
NH	1	1	(i)		0.995	0.950	0.958	0.755	0.529	0.334
NH	8	2	(i)		0.860	0.558	0.310	0.207	0.124	0.108
NH	17	2	(i)		1.110	1.360	1.350	0.911	0.904	0.732
NH	18	2	(i)		0.884	0.791	0.892	0.780	0.146	0.054
										0.045
NI	5	2	(i)		0.632	0.434	0.194	0.106	0.043	0.120
NI	9	2	(i)		1.050	0.520	1.190	0.770	0.190	0.217
								0.733	0.281	0.206
NJ	4	1	(i)		0.805	0.630	0.242	0.107	0.049	0.082
										0.044
NJ	12	1	(ii)		1.160	1.300	1.260	0.830	0.318	0.136
NJ	12	2	(i)		0.553	0.830	1.060	0.780	0.312	0.110
							1.060			
NJ	15	1	(i)		1.010	1.020	0.985	0.680	0.203	0.065

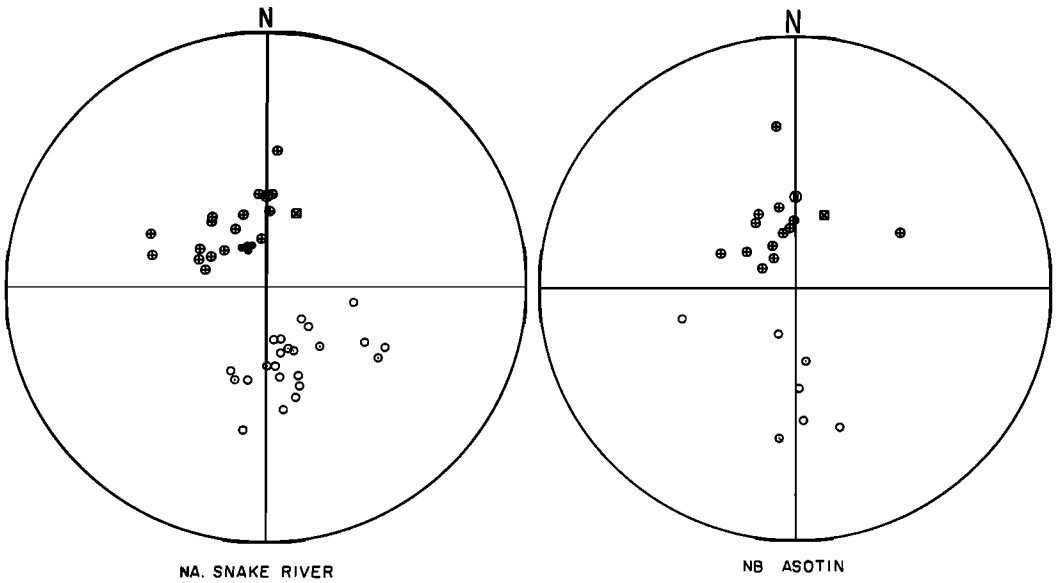


Fig. 14. Directions of NRM in all cores from sections *NA* and *NB* after treatment in an alternating magnetic field of 200 oe. Symbols as for Figure 4.

ferent polarity epoch are evident in the south. Very strong evidence of a rapid extrusion rate in the southern area has been revealed in a separate study of the NRM in a very large number of lavas well below the summit of Steens Mountain [*Watkins*, 1965]. A high-density sampling survey has yielded data which include a well-defined transition zone from N to R polarities across 19 lavas. No details of the data are included here, but the mean direction of NRM in each lava concerned is given in Figure 20. *Cox et al.* [1964, p. 1542] reason that such

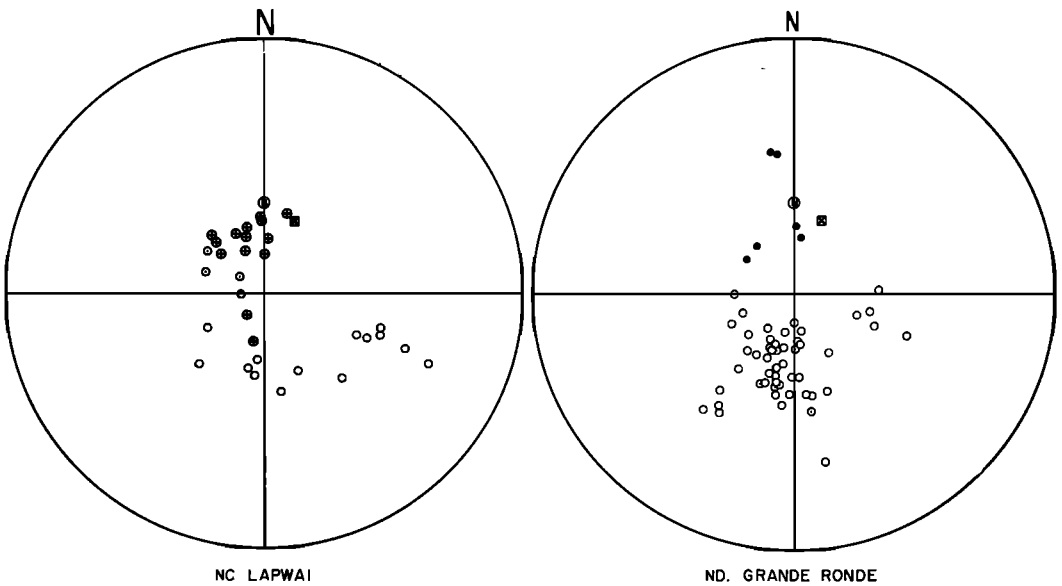


Fig. 15. Directions of NRM in all cores from sections *NC* and *ND* after treatment in an alternating magnetic field of 200 oe. Symbols as for Figure 4.

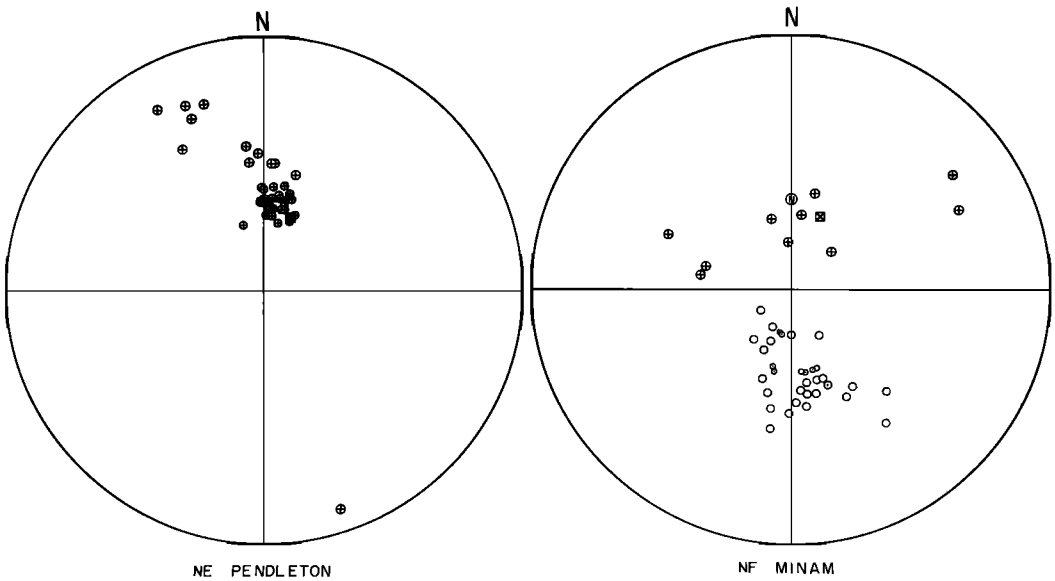


Fig. 16. Directions of NRM in all cores from sections *E* and *F* after treatment in an alternating magnetic field of 200 oe. Symbols as for Figure 4.

transitions, which have been rarely observed, may last only 10,000 years, in which case the extrusion rate of Steens basalt might be of the order of an average of one lava every 500 years for the transition period.

Successive polarity variation between groups of lavas in sections of the Columbia River basalt is common enough to be considered a potentially valuable stratigraphic tool. On the other hand, in the southern area the lack of polarity variation in large sections is correspondingly discouraging. *Watkins* [1963*b*] has shown, however, that apparently systematic variations of NRM within single polarity zones of lavas in

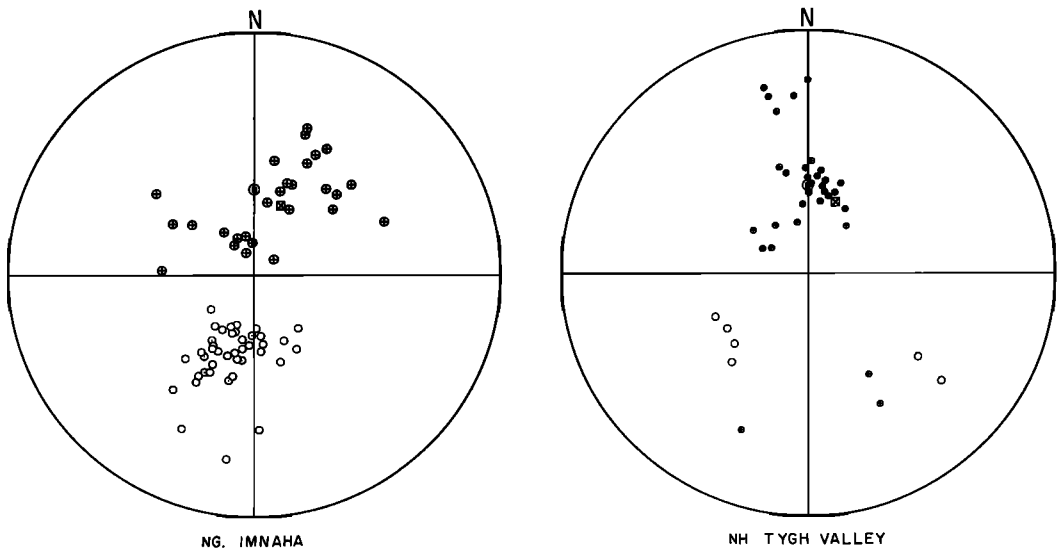


Fig. 17. Directions of NRM in all cores from sections *G* and *H* after treatment in an alternating magnetic field of 200 oe. Symbols as for Figure 4.

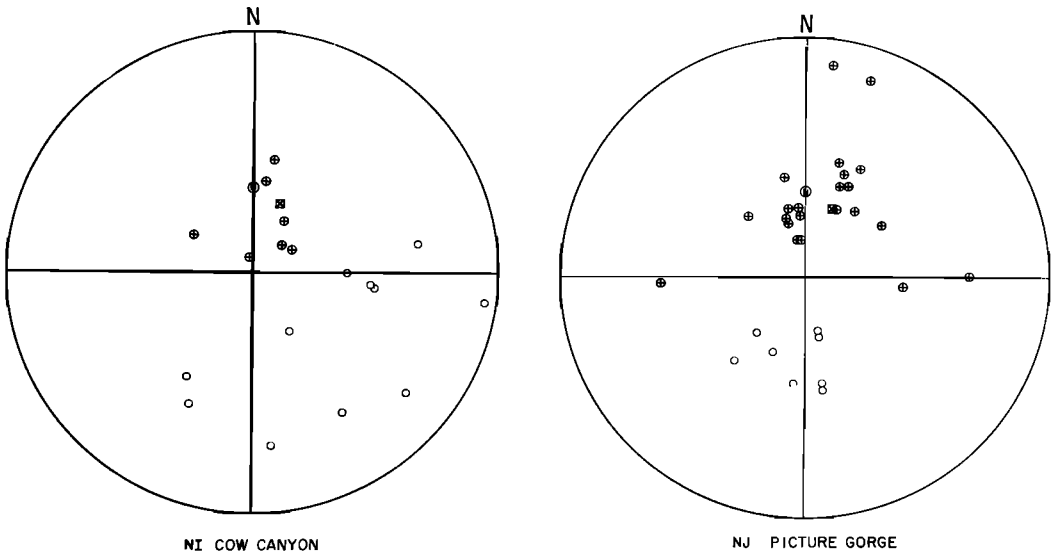


Fig. 18. Directions of NRM in all cores from sections *I* and *J* after treatment in an alternating magnetic field of 200 oe. Symbols as for Figure 4.

southern Oregon may be revealed with high-density sampling, so that the NRM in the southern area may still be regarded as a potentially useful stratigraphic tool.

The alternating nature of the polarity of the geomagnetic field with time [*Cox et al.*, 1963] is evidence that use of magnetic polarity as an independent positive correlation tool over long distances is dangerous. *Cox et al.* [1964] have shown that when integrated with an independent method such as K-Ar dating, magnetic polarities serve to confirm and define correlations on a global scale. As a first approximation, it may seem possible that the lavas of normal polarity in Picture Gorge (section *NJ*) were extruded during the same magnetic polarity epoch as the lavas of normal polarity examined from the top of Steens Mountain (section *SJ*), since *Evernden and James* [1964] have dated rocks immediately above the respective sections as 15.4 and 14.5 m.y. old. The durations of the polarity epochs in the Miocene period have not

TABLE 6. Columbia River Basalts Statistical Analysis of Results for Sections *NA* to *NJ*

Section No.	D	I	R	N	B	α_{95}, deg (1 level)	w	β	α_{95}, deg (2 level)	δ_p, deg (1 level)	δ_m, deg (1 level)	k	ϕ'	θ'
NA	337°07′	+66°17′	21.222	44	22	6.1	136.98	38.64	5.1	8.2	10.0	27.0	168°35′	71°50′
NB	334°29′	+66°20′	9.136	20	10	15.7	7.77	24.62	14.4	21.2	25.0	10.4	145°23′	73°21′
NC	339°08′	+73°04′	15.084	32	16	9.2	99.21	18.04	8.6	15.0	16.8	16.4	199°49′	72°25′
ND	1°57′	+67°47′	28.533	60	30	6.0	33.24	27.13	5.8	8.3	10.2	19.7	259°12′	85°35′
NE	359°01′	+54°04′	17.132	36	18	8.0	177.00	23.92	7.0	7.9	11.2	21.0	56°51′	78°57′
NF	0°05′	+68°01′	20.922	44	22	8.6	64.35	23.21	6.7	12.4	14.6	13.0	242°40′	83°05′
NG	12°45′	+61°43′	33.284	72	36	6.9	34.94	20.85	5.8	8.2	10.7	19.8	349°51′	80°14′
NH	359°54′	+61°42′	17.015	38	19	11.8	65.62	12.32	9.6	14.1	18.2	9.0	56°05′	87°36′
NI	314°05′	+71°36′	7.29	20	10	31.4	10.78	6.29	20.1	38.4	55.2	3.3	185°28′	59°21′
NJ	13°50′	+62°03′	14.08	30	15	10.3	56.88	14.19	10.1	12.6	16.2	15.2	330°23′	79°59′
All sections	356°17′	+64°51′	182.2	396	198	2.9				3.8	4.8	12.4	171°44′	87°06′

All data from magnetically reversed lavas are transposed to northern hemisphere.
Symbols are same as those in Table 3.

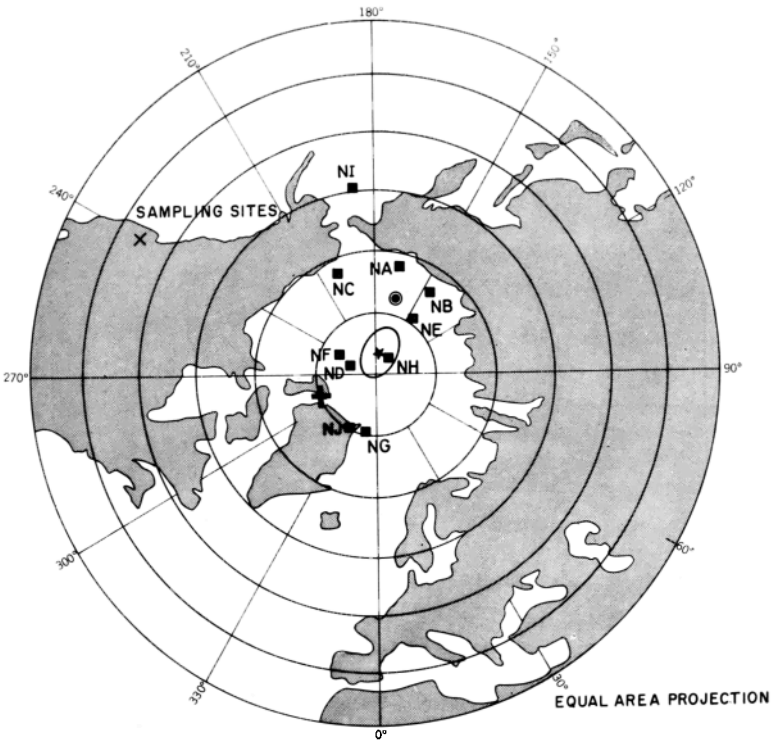


Fig. 19. Virtual geomagnetic poles for sections *NA* to *NJ*. Symbols as in Figure 10.

yet been defined, however; and recent work by *Cox et al.* [1964] shows that the polarity epochs so far determined for younger rocks are not necessarily of even approximately equal durations. The existence of 'events' [*Cox et al.*, 1964, p. 1542], or short-period polarity epochs, in younger rocks suggests the possibility of further difficulties in positive long-range correlation studies using magnetic polarities, unless the data are integrated with independent methods.

The specimens taken from the baked interbasaltic zones all have NRM with the same polarity as that in the respective overlying lavas. This strongly suggests that in these particular cases the source of the observed polarities, which are all reversed, is likely to be the geomagnetic field and not a self-reversing mechanism such as that described by *Uyeda* [1958].

Dual direction of NRM. Unusual dual directions of NRM occur in sections *SD* and *SH*, but not in any northern section. *Athavale et al.* [1963] have described dual directions of NRM in Indian Jurassic rocks but have attempted no explanation for this phenomenon. The dual directions in sections *SD* and *SH* are stratigraphically separated but are immediately conformable, and thus may be explained by one of the following hypotheses:

(1) The lavas were extruded at a rapid rate during normal secular variation of the geomagnetic field. A pause in extrusion during this movement of the magnetic field could create the observed dual distribution of NRM.

(2) The lavas were extruded at a regular rate, and the dual direction is due to a relatively sudden movement of the ancient ambient magnetic field.

(3) Tectonic activity may be suspected to have caused the dual distribution of NRM. The dual distribution in section *SH*, however, is certainly not a result of tectonic rotation during the period between extrusion of the critical lavas, superimposed on a relatively unchanging ambient magnetic field. The hypothetical rotation would need to be about an inclined axis, and all lavas in the section are flat. The dual direction in section *SD* might be suspected to be

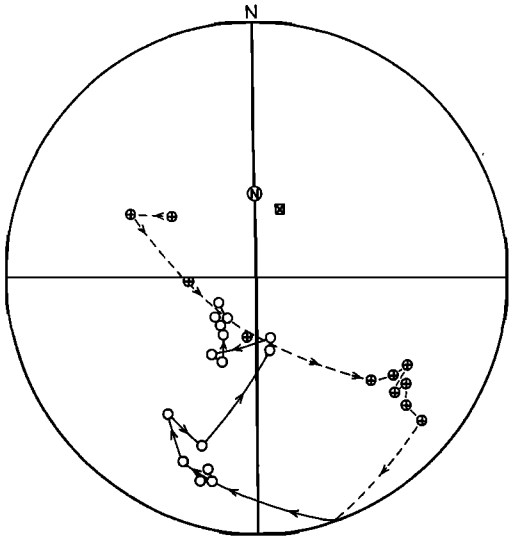


Fig. 20. Directions of NRM in a sequence of lavas approximately 450 m below the summit of Steens Mountain (section *SJ*). Each direction is the mean NRM in a single lava, derived from at least six orientated cores, after treatment of each core in an alternating magnetic field of 200 oe. Symbols as for Figure 4. In addition, NRM for successive lavas are joined by a dashed line on lower hemisphere and a full line on upper hemisphere.

a function of structural rotation in a direction corresponding to the orocline or gigantic structural feature (Figures 1 and 11, insets) which *Carey* [1958] and others suspect may dominate the tectonic pattern of the cordillera. The relative ages of the groups, indicated in Figure 5, are such, however, that the implied rotation is in the wrong direction to explain the dual direction of NRM.

The first hypothesis is slightly more favored by present evidence. As argued in the discussion of polarities, a rapid lava extrusion rate is probable in the southern area. In addition, the polarity transition zone [*Watkins*, 1965] which was mentioned earlier and is illustrated in Figure 20 tends to feature well-defined groupings of NRM rather than a regular variation of NRM direction between the N to R lavas above and below the transition. Although it seems more feasible to attribute these groupings to a pulsing of volcanic activity superimposed on the movement of the ambient magnetic field from N to R polarity rather than to sudden movements of the geomagnetic field during the implied transition period superimposed on regular lava extrusions, the latter possibility cannot be altogether dismissed.

The virtual geomagnetic poles for each section. It has become conventional to represent the measured NRM in terms of the virtual geomagnetic pole, or surface expression of the geocentric dipole, which would cause the observed mean magnetic directions. *Runcorn* [1959] suggested that a period of about 10,000 years may be required to average nonaxial dipole components to the spin axis. *Currie et al.* [1963, p. 2277] consider that a period of up to 1 million years may have been needed to accomplish the same condition during the Cretaceous period in California. The virtual geomagnetic poles presented in Figures 10 and 19 are not coincident with each other. Although the 95% confidence ovals have been omitted for clarity, many of the ovals, if plotted, would not overlap. The cause of this lack of coincidence of virtual geomagnetic poles is likely to be one of three possibilities which are discussed below.

(1) It is considered highly probable that the more rapidly moving nondipole sources [*Bullard et al.*, 1950] would average out to the inclined dipole in most sections, assuming that such behavior and terminology [*Cox and Doell*, 1960, pp. 662–664] are applicable to the Miocene geomagnetic field. It is possible, however, that insufficient duration is represented in some of the sections sampled to average the inclined dipole to an axial dipole state and that this causes the observed distribution of virtual geomagnetic poles. Since *Doell and Cox* [1961*b*] have shown that thick sections of lavas can accumulate in a time which is insufficient to average nondipole components to the axial dipole in Hawaii, it is not impossible that the virtual geomagnetic poles shown in Figures 10 and 19 represent inclined dipoles. The previously presented polarity data and discussion suggest that if any of the virtual geomagnetic poles represent inclined dipoles they are for samples from the southern area. The polarity data and limited absolute age determinations [*Evernden and James*, 1964] for the northern area, particularly for the sections possessing markedly different associated virtual geomagnetic pole positions close to the Oregon-Washington border, imply time spans for the sections of the order of at least 5 million years.

If inclined dipoles persist for as long as 5 million years, a considerably longer period would be required to average the inclined dipole to the spin axis.

(2) It may be argued that the virtual geomagnetic poles in Figures 10 and 19 are actually expressions of axial dipoles, but if the sections are not all contemporaneous the distribution of the virtual geomagnetic poles could be caused by relatively rapid movement of the spin axis and associated axial dipoles with respect to the crust of the earth. Polar wandering, as such, seems to be rejected by most of the studies of large amounts of paleomagnetic data [*Blackett et al.*, 1960], and it is doubtful that the critical sections are so absolutely uncontemporaneous as would be necessary for this possibility to be a fact.

(3) Explanations of broadly contemporaneous noncoincident virtual geomagnetic poles in terms of tectonic modification of the sections or areas concerned are common in the literature. It is convenient to examine the possibility of a systematic modification of the NRM of the Columbia plateaus by relative tectonic rotation by examining a diagram of the mean paleomagnetic declination in each section (Figure 21). The mean paleomagnetic declination in a section of Eocene lavas in the Siletz River area [*Cox*, 1957] has been added to the diagram. *Irving* [1959 and 1964, p. 249] has commented on this Eocene result, suggesting that the associated virtual geomagnetic pole, which is well removed from results based on studies of Eocene rocks in other parts of the world, may be anomalous because of tectonic rotation in this tectonically still-active area. *Van Hilten* [1964, p. 50] also prefers to explain *Cox*'s [1957] result in terms of broad-scale tectonic movement. To suppress the data from sections with poorly defined mean NRM directions, the declination symbols in Figure 21 are proportional in length to the logarithm of the precision parameter k (Table 3). In section SH_2, however, it must be realized that the high-precision parameter is fortuitous, being due largely to the fact that only four lavas were sampled. This condition is emphasized by the magnitude of the between-lava confidence parameter (β) in the section. A further difference in the precision of data must be expected from the probable difference in the time span of sections in the two surveys.

It is frequently valuable to compare paleomagnetic data with similar data from other continents. *Doell and Cox* [1961*a*, p. 294], using the results from several relatively reliable surveys of paleomagnetism in Miocene rocks elsewhere in the world, propose a mean virtual geomagnetic pole in the vicinity of longitude 165°E, latitude 77°N. This implied position for the Miocene geomagnetic pole, or spin axis, may well be modified with the results from more surveys or precise knowledge of continental displacements since the Miocene period, but for the sake of comparison the magnetic meridians for the above virtual geomagnetic pole are included in Figure 21. This mean virtual geomagnetic pole is also included in Figures 10 and 19.

If the mean paleomagnetic declinations in each section represent axially dipolar fields and are not parallel with the magnetic meridians of the mean Miocene virtual geomagnetic pole discussed above, it is likely that tectonic rotation has taken place. Figure 21 exhibits a striking relationship between the mean paleomagnetic declinations and the orocline, particularly in the data from the northern survey, which has previously been argued to represent longer time spans than the data from the south. Much of the declination data, then, is consistent with a right-lateral rotation of the orocline. Two points are worth emphasis. First, sections *NA*, *NB*, and *NC* have paleomagnetic declinations suggesting that no rotation has taken place in their particular vicinity. This is what could be expected in the case of right-lateral rotation, which *Wise* [1963] suggests may have taken place between the relatively stably orientated Olympic-Wallowa lineament and the Walker lineament. *Carey* [1958] suggested a similar movement, but between the Rocky Mountain trench and the San Andreas fault. Second, the Eocene paleomagnetic declination measurement by *Cox* [1957] is consistent with this hypothesis; he inferred a rotation, much of which must have taken place during the early Tertiary, after the formation of the Franciscan basement. With continued tectonic activity since the Early Tertiary, the observed paleomagnetic declination rotation would be a function of age.

Several of the paleomagnetic declinations in Figure 21 do not support the orocline hypothesis. This condition could be due to one or both of two main causes. The sections concerned

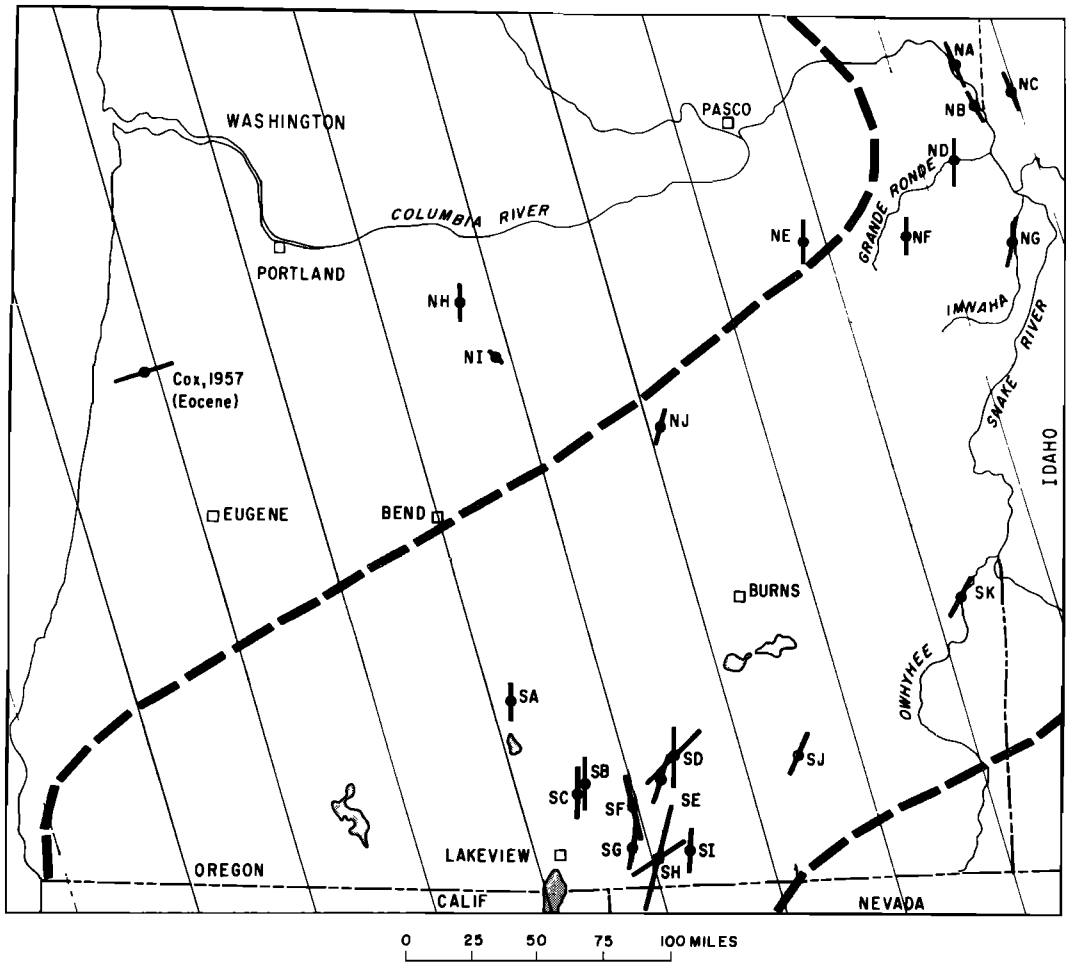


Fig. 21. Mean paleomagnetic declinations in all sections with Miocene magnetic meridians, and inferred orocline. Magnetic meridians are due to virtual geomagnetic pole at 165°E, 77°N [*Doell and Cox,* 1961*a*, p. 294]. Mean paleomagnetic declination is represented by short line at each section. Miocene magnetic meridians represented by fine continuous lines. Inferred orocline margin represented by heavy dashed line.

may not represent a sufficient length of time for the average field to approximate the axial dipole. As previously argued, this is more likely to occur in sections in the southern survey, where, in addition, the location is such that the effect of the rotation being examined would be less than in the north. The southern survey may also be subjected to very local tectonic modification. *Donath* [1962] has described the great amount of normal faulting in the area close to sections *SA, SB,* and *SC*. The second main cause of recalcitrant paleomagnetic declinations would be the fact that some sections may be much younger than others, in which case the observed rotation would be less for those sections.

Since it has been shown that in the southern area no tectonic forces are necessary to produce in single sections two different mean directions of NRM, it might be imagined that the same argument could be applied to the variation of NRM between sections and thus that the relation of all the data to the orocline, as illustrated in Figure 21, is accidental. The possibility of such a coincidence must be remote indeed. It is difficult to accept the correlation between an independently proposed tectonic pattern and

paleomagnetic data in the Columbia plateaus as anything but cause and effect. Further tests of the validity of the paleomagnetic support for the orocline hypothesis can be made. Ideally, measurement of the paleomagnetic declination in a single borehole in north-central Oregon would test the required rotation as a function of time. Since this is impractical, extension of the paleomagnetic survey to older sections in the Columbia plateaus and accurate dating of the sections concerned would assist in resolving the remotely possible ambiguity outlined above.

The mean virtual geomagnetic pole for the entire survey. Figures 10 and 19 illustrate the lack of coincidence of the mean virtual geomagnetic poles for the two surveys. This is consistent with the preferred interpretation of tectonic modification of the NRM directions.

The NRM in both surveys has been combined and statistically analyzed according to the methods of *Fisher* [1953]; unit magnitude was assigned to the vector for each lava. The results are presented in Table 7.

TABLE 7. Statistical Analysis of Combined Data from the Two Surveys

ϕ	119°37′
θ	44°31′
D	2°56′
I	+62°47′
R	399.37
N	911
B	433
α_{95}	6.4°
δ_p	7.3°
δ_m	10.5°
k	12.8
ϕ'	25°40′
θ'	86°01′

Symbols are same as those in Table 3.

Doell and Cox [1961*a*, p. 293] have pointed out that the Miocene paleomagnetic data so far available suggest that the spin axis was removed from its present position during the Miocene period. The oval of confidence of the mean virtual geomagnetic pole encloses the present spin axis, but, since the data which determined this confidence oval are derived from lavas which are likely to have undergone tectonic rotation since cooling, little significance can be attached to the position of the mean virtual geomagnetic pole for the entire survey, in terms of the implied position of the Miocene spin axis.

Conclusion

The paleomagnetism in Miocene lavas in the Columbia plateaus is largely consistent with *Carey*'s [1958] orocline hypothesis for the tectonic pattern of the cordillera and is some support for *Bodvarsson and Walker*'s [1964] suggestion that tectonic spreading may be a feature of the Columbia plateaus. Much of the tectonic movement of the orocline is likely to have taken place since the Miocene period.

This conclusion can be tested further.

The paleomagnetic data most reasonably should be added to much independent geological, geophysical, and geochemical evidence, summarized by *Wise* [1963] to support a tectonic origin for the broad structural pattern of the cordillera. If the conclusion were to be rejected, among the alternatives involved in explaining the data it would be necessary to propose complete noncontemporaneity of large sections of lavas fairly close to each other near the Oregon-Washington border and to invoke the persistence of inclined dipole conditions for periods probably well in excess of 5 million years.

These two surveys have added information concerning contrasts between the magnetic stability, polarity variation, implied lava extrusion rates, and stratigraphic utility of field measurements of magnetic polarity in the lavas to the already observed contrasts in physiography, local tectonic features, and petrology between those parts of the Columbia plateaus north and south of the Ochoco–Blue Mountains axis.

The results described in this paper serve to emphasize the danger in attempting to define the position of the spin axis from paleomagnetic data for a single section of rocks. The possibility, however remote, of a regional tectonic change of paleomagnetic directions must be examined before conclusions can be drawn concerning the positions of the ancient axial dipole or spin axis.

Acknowledgments. Acknowledgments are due Drs. A. Cox and R. Doell of the U. S. Geological Survey for generous loan of equipment; Dr. R. Doell for review of the manuscript; G. W. Walker of the U. S. Geological Survey for advice on the geology of the southern area; Professor A. C. Waters of the University of California, Santa Barbara, for advising in the selection of suitable sections in the northern area; and Dr. R. L. Wilson of the University of Liverpool for review of the manuscript.

The survey described in part 1 was supported by U. S. Air Force contract AF 19(604)-8043, and constituted the basis of a thesis for the degree of Ph.D. of the University of London. The survey described in part 2 was supported by a National Science Foundation grant to Stanford University, as part of the United States–Japan Cooperation Program.

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(Manuscript received June 8, 1964;
revised December 4, 1964.)